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Finite-Time Adaptive Dynamic Surface Control of a Quadrotor with Input Delay

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Abstract
This paper presents finite-time adaptive multi-dimensional Taylor network (MTN)-based dynamic surface control (DSC) for attitude and position control of a quadrotor subjected to input delay, parametric uncertainties, and external disturbances. The input delay is compensated using the Pade approximation approach. Due to the simple structure, low computation, and excellent approximation capability of the MTN, it is employed to approximate the uncertain nonlinearities of the quadrotor system. The MTN is updated using a single parameter instead of a vector of many weights to simplify it further. An improved DSC strategy is employed to achieve the attitude and position tracking of the quadrotor in finite time. Different from the traditional first-order filter of the DSC, a modified first-order filter is proposed such that it not only prevents the "explosion of complexity" problem associated with the conventional backstepping approach but also utilizes extra terms to guarantee finite-time convergence and estimates and compensates the problem of boundary layer error. Thus, the assumption that the maximum bound of the derivative of the virtual control law is known is no longer required. The gains of the DSC are updated online so that proper gains are adapted in response to the change in the reference trajectories to improve the tracking accuracy. Lyapunov candidate function is used to prove the finite-time convergence of the closed-loop system to a compact set. Finally, numerical simulations have been provided to demonstrate the advantages of the improved DSC scheme.

Keywords Quadrotor · Sliding mode control · Backstepping · Delay · Multi-dimensional Taylor network · Dynamic surface control

1 Introduction

Quadrotors belong to the class of unmanned aerial vehicles (UAV) extensively used in inspection, mapping, rescue mission, maritime cruising, military surveillance, aerial cinematography, and so on [1, 2]. They are lighter in weight, have a simple structure, capable of vertical take-up and landing, hovering, and performing rapid maneuvering with greater agility [3].

In recent years, quadrotors have received great attention from researchers with a view to enhancing their operations via control. However, quadrotors pose serious challenges due to their strongly coupled nonlinear dynamics, multi-input multi-output, unmodeled dynamics, time-varying parame-

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ters, and susceptibility to disturbances [4]. Furthermore, the quadrotors are underactuated systems that have six degrees of freedom with just four actuators [5, 6]. Hence, it is difficult to control the coupled nonlinear and underactuated quadrotor UAV.

Various control methods have been put forward to overcome the challenges. In [7], a robust approach using a fixed-time sliding mode control has been designed for a quadrotor aircraft. An adaptive sliding mode controller with improved disturbance suppression capability has been implemented for a quadrotor in [8, 9]. In [10], a sliding mode control method has been presented to tackle complex quadrotor control situations by identifying fault vectors and computing suitable control inputs. In [11], a novel backstepping-based fault-tolerant control system has been developed and tested on a quadrotor UAV testbed in the presence of actuator faults. Maaruf et al. [12] studied the trajectory tracking problem of a quadrotor using an adaptive robust sliding mode and backstepping control. In [13], the quadrotor control design improves the tracking accuracy and robustness of the proposed integrated backstepping sliding mode control. In [14], the study provides a backstepping sliding mode control for the attitude tracking of a quadrotor in the presence of modeling errors and wind perturbations.

Recently, the application of dynamic surface control (DSC) for trajectory tracking of quadrotors is attracting the attention of researchers. The DSC utilizes a low-pass filter to estimate the derivative of the virtual control law to prevent the "explosion of complexity" associated with the traditional backstepping control strategy. In [15], the authors proposed DSC and adaptive dynamic programming to achieve the trajectory control of a quadrotor. In [16], an observer-based DSC of a quadrotor with external disturbances has been studied. In [17], an adaptive DSC was designed for a quadrotor with parametric uncertainties and a slung load. In [18, 19], an adaptive DSC with a neural network was designed for a quadrotor subjected to parametric uncertainties. In [20], an adaptive neural observer-based DSC of a quadrotor with input hysteresis has been studied. Considering the low volume of DSC strategies applied to the quadrotors recently, it is still open for more investigations to enhance its performances which is the motivation of this work.

In the flight control system of quadrotors, time delays arise in the control inputs as a result of latency of computation and communication, and aging [21]. Numerous studies have been undertaken to address the control challenges faced by quadrotor systems that are subjected to time delays. In [22], a model reference adaptive control was proposed for the linearized dynamic model of the quadrotor. In [23], a nonlinear dynamics inversion control integrated with a lead compensator was designed for a quadrotor with input delay. In [24], backstepping control with a nonlinear disturbance observer was proposed for a quadrotor with input delay and external disturbances. In [25], a barrier Lyapunov function-based backstepping control has been utilized to control the vertical position of a quadrotor with input delay. In [26], the authors designed an integral backstepping control for position and rotation control of a quadrotor with input delay. In [27], a fractional-order backstepping control of a quadrotor with input delay was investigated. It should be noted that these control methods are based on asymptotic stability. This implies that the quadrotors subjected to time delay achieve trajectory tracking as the time approaches infinity. This could compromise the tracking accuracy of the quadrotors and is undesirable. Thus, how to design a controller with guaranteed convergence within a short finite time for a quadrotor with a time-delay issue is another motivation for this work.

Finite-time control strategies guarantee the attainment of the control objectives with finite-time stability contrary to asymptotic stability which guarantees the realization of the control objectives as time approaches infinity [28]. Finite-time control strategies offer some advantages such as improved robustness, rapid convergence to the target trajectories, and enhanced tracking accuracy. Several finite-time control techniques have been reported for quadrotors [29–34]. For example, the authors in [28] introduce an adaptive finite-time backstepping control technique enabling a quadrotor to follow specified trajectories accurately. In [35], a finite-time control of a quadrotor with state constraints has been studied. An adaptive finite-time backstepping approach has been used for the trajectory tracking of a quadrotor with external disturbances in [28, 36]. In [37], a finite-time backstepping sliding mode control has been designed for a quadrotor with parametric uncertainties. In [38], a global sliding mode control strategy with finite-time convergence property has been employed for trajectory tracking of an uncertain quadrotor. In [39], an adaptive finite-time fractional-order sliding mode control of a quadrotor with unknown parameters has been studied. It is obvious that there exist a plethora of finite-time sliding mode control and finite-time backstepping control techniques applied to the quadrotors, but a finite-time DSC of a quadrotor is not widely discussed.

The first-order filters in the existing DSC strategies applied to quadrotors [15–20] have the problem of boundary layer errors. The boundary layer errors lead to a large tracking error bound, which affects the control accuracy. Therefore, it is essential to modify the first-order filter by incorporating more terms so that it not only prevents the problem of "explosion of complexity" but also compensates for the effect of the boundary layer errors without requiring any knowledge about their upper bounds, thereby enhancing the tracking accuracy of the DSC. The existing DSC strategies and their corresponding first-order filters [15–20] are based on asymptotic convergence which prompts the need to improve their convergence speed in the presence of environmental disturbances

using the finite-time stability theorem. Moreover, the gains of the DSC [15–20] are fixed for all maneuvers. However, considering that environmental disturbances are unpredictable and quadrotors can perform high and low maneuvers, using fixed gains under these conditions may lead to unsatisfactory control performance. Therefore, the controller gains should be adaptive so that they can automatically adjust accordingly in response to changes in target maneuvers and environmental disturbances to improve robustness. In addition, the DSC strategies [15, 18–20] utilize neural networks to estimate the uncertain nonlinearities. Nonetheless, most neural networks have major drawbacks such as complex structure, high computational requirements, low efficiency, and long training period, which may give rise to poor real-time performance. Thus, it is essential to employ a simple but effective approximation technique, such as the multi-dimensional Taylor network (MTN) to replace the neural networks. The MTN has some critical characteristics such as simple structure, few parameters, fast convergence speed, and excellent approximation ability [40, 41]. Meanwhile, the use of the MTN to approximate the uncertain nonlinearities of a quadrotor is rare. Therefore, it is of utmost importance to study the finite-time DSC with an adaptive MTN and a modified first-order filter to improve the trajectory tracking accuracy of the quadrotor with input delay.

Motivated by the literature discussions, this paper proposes an MTN-based improved DSC with adjustable gains and a modified first-order filter to guarantee the convergence of the quadrotor with input delay in a short finite time. Addressing these problems simultaneously is difficult and has not been investigated before. The challenge brought by the input delay is addressed using the Pade approximation. The next challenge uses the MTN to approximate the unknown parameters and the external disturbances of the quadrotor. The MTN is updated using a single adaptive parameter to lower its computational cost. Another challenge in the control design is modifying the first-order filter of the DSC with extra terms to tackle not only the "explosion of complexity" but also to mitigate the effect caused by the boundary layer error using adaptive laws, and guarantee finite-time convergence. Thus, the prior knowledge of the upper bound of the boundary layer error is not required. The gains of the DSC are updated online so they can self-adjust whenever the target trajectories change. The proof process to demonstrate the finite-time boundedness of the closed-loop system under the proposed control is tedious. The important contributions of the proposed control scheme are summarized as follows:

1. This is the first paper to investigate an adaptive finite-time MTN-based improved DSC with a modified first-order filter for the trajectory tracking of a quadrotor with input delay. The Pade approximation approach is used to mitigate the input delay.
2. Different from the widely employed neural network approximations for adaptive control of quadrotors [15, 18–20], this paper uses the MTN-based approximation for the quadrotor. Contrary to neural network approximations, the MTN is easier to implement in practice and has a simple structure with polynomials in the hidden layer, fast convergence, and excellent approximation capability. To further simplify the MTN and reduce the computational cost, a single adaptive parameter is updated instead of the weight vector.
3. Unlike the DSC approaches in [15–20], the first-order filter of the DSC is modified with additional terms so that it adaptively mitigates the problem of boundary layer error and ensures convergence in finite time. Thus, prior information about the upper bounds of the derivatives of virtual control law is no longer required. This improves the estimation accuracy of the low-pass filter and the accuracy of the trajectory tracking. Furthermore, adaptive laws are used to adjust the DSC gains online whenever the desired trajectories change. This allows the DSC to regulate the quadrotor with the appropriate gains for every operation.

The rest of the article is organized as follows. In Sect. 2, the nonlinear mathematical model of the quadrotor is presented. In Sect. 3, the design steps of the enhanced DSC strategy are presented. The simulation results and performance comparison with existing techniques are discussed in Sect. 4. Some concluding remarks are provided in Sect. 5.

2 Mathematical Modeling

The schematic diagram of the quadrotor is presented in Fig. 1. The description as well as the mathematical modeling of quadrotors has been extensively studied in the literature[13, 24]. The state space dynamic equations describing the six-degree-of-freedom quadrotor are formulated based on the standard quadrotor model in [24], extended to incorporate time-delay effects following the approach in [13] as:

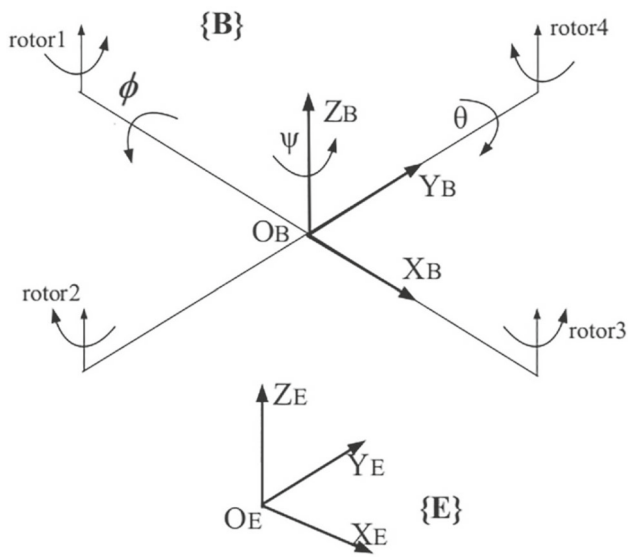

Fig. 1 Schematic diagram of the quadrotor [24]

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = \mathrm{a}_1 x_4 x_6 + \mathrm{a}_2 x_4 + \mathrm{a}_3 x_2^2 + b_1 U_1(t-\iota) \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = \mathrm{a}_4 x_2 x_6 + \mathrm{a}_5 x_2 + \mathrm{a}_6 x_4^2 + b_2 U_2(t-\iota) \\ \dot{x}_5 = x_6 \\ \dot{x}_6 = \mathrm{a}_7 x_2 x_4 + \mathrm{a}_8 x_6^2 + b_3 U_3(t-\iota) \\ \dot{x}_7 = x_8 \\ \dot{x}_8 = \mathrm{a}_9 x_8 + (C_{x_2} C_{x_5} S_{x_3} + S_{x_2} S_{x_5}) \frac{\bar{U}_4(t-\iota)}{m} \\ \dot{x}_9 = x_{10} \\ \dot{x}_9 = \mathrm{a}_{10} x_{10} + (C_{x_2} S_{x_3} S_{x_5} + C_{x_5} S_{x_2}) \frac{\bar{U}_4(t-\iota)}{m} \\ \dot{x}_{11} = x_{12} \\ \dot{x}_{12} = -g + \mathrm{a}_{11} x_{12} + (C_{x_2} C_{x_3}) \frac{\bar{U}_4(t-\iota)}{m} \end{cases} \tag{1}$$

where x_1, x_3, and x_5 denote the roll, pitch, and yaw orientations, respectively, x_7, x_9, and x_{11} indicate the quadrotor position. $\mathrm{a}_1 = \frac{J_y - J_z}{J_x}$, $\mathrm{a}_2 = \frac{\Omega_r J_r}{J_x}$, $\mathrm{a}_3 = \frac{-K_{a\phi}}{J_x}$, $\mathrm{a}_4 = \frac{J_z - J_x}{J_y}$, $\mathrm{a}_5 = \frac{\Omega_r J_r}{J_y}$, $\mathrm{a}_6 = \frac{-K_{a\theta}}{J_y}$, $\mathrm{a}_7 = \frac{J_x - J_y}{J_i}$, $\mathrm{a}_8 = \frac{-K_{a\psi}}{J_z}$, $\mathrm{a}_9 = \frac{-K_{dx}}{m}$, $\mathrm{a}_{10} = \frac{-K_{dy}}{m}$, $\mathrm{a}_{11} = \frac{-K_{dz}}{m}$, $b_1 = \frac{L}{J_x}$, $b_2 = \frac{L}{J_y}$, $b_3 = \frac{L}{J_z}$, m is total quadrotor mass, g denote the gravity constant, J_x, J_y, J_z represent the quadrotor moment of inertia, K_{ax}, K_{ay}, K_{az}, $K_{a\phi}$, $K_{a\theta}$, and $K_{a\psi}$ are the drag constants, U_1, U_2, and U_3 are the roll, pitch, and yaw inputs, respectively, $\bar{U}_4$ is the altitude control signal, and K_p and C_d are the aerodynamic constants.

The false inputs are defined by [24]

$$\begin{cases} U_4(t-\iota) = (S_{x_2} S_{x_5} + S_{x_3} C_{x_2} C_{x_5}) \frac{\bar{U}_4(t-\iota)}{m} \\ U_5(t-\iota) = (S_{x_2} C_{x_5} + S_{x_3} S_{x_5} C_{x_2}) \frac{\bar{U}_4(t-\iota)}{m} \\ U_6(t-\iota) = (C_{x_2} C_{x_3}) \frac{\bar{U}_4(t-\iota)}{m} \\ \bar{U}_4(t-\iota) = m\sqrt{U_4^2(t-\iota) + U_5^2(t-\iota) + U_6^2(t-\iota)} \end{cases} \tag{2}$$

The following equation is obtained by substituting (2) into (1).

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f_1 + b_1 U_1(t-\iota) \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = f_2 + b_2 U_2(t-\iota) \\ \dot{x}_5 = x_6 \\ \dot{x}_6 = f_3 + b_3 U_3(t-\iota) \\ \dot{x}_7 = x_8 \\ \dot{x}_8 = f_4 + b_4 U_4(t-\iota) \\ \dot{x}_9 = x_{10} \\ \dot{x}_9 = f_5 + b_5 U_4(t-\iota) \\ \dot{x}_{11} = x_{12} \\ \dot{x}_{12} = f_6 + b_6 U_4(t-\iota) \end{cases} \tag{3}$$

where

$$f_1 = \mathrm{a}_1 x_4 x_6 + \mathrm{a}_2 x_4 + \mathrm{a}_3 x_2^2, \;\; f_2 = \mathrm{a}_4 x_2 x_6 + \mathrm{a}_5 x_2 + \mathrm{a}_6 x_4^2$$
$$f_3 = \mathrm{a}_7 x_2 x_4 + \mathrm{a}_8 x_6^2, \;\; f_4 = \mathrm{a}_9 x_8, \;\; f_5 = \mathrm{a}_{10} x_{10},$$
$$f_6 = a_{11} x_{12} - g b_4 = 1, \;\; b_5 = 1, \;\; b_6 = 1$$

For the purpose of control design, (3) can be simplified as follows:

$$\begin{cases} \dot{x}_{2i-1} = x_{2i} \\ \dot{x}_{2i} = f_i + b_i U_i(t-\iota) \quad (i = 1, 2, 3, 4, 5, 6) \end{cases} \tag{4}$$

The uncertain nonlinear functions f_i $(i = 1, 2, 3, 4, 5, 6)$ will be approximated with the aid of an MTN updated via adaptive learning. To compensate for the input delays in (4), the Laplace transform and Pade approximation are introduced. Considering the Laplace transform and delay theorem, one gets [42, 43]:

$$\mathcal{L}\{U_i(t-\iota)\} = \exp(-\iota s)\mathcal{L}(U_i(t)) = \frac{\exp(-\iota s/2)}{\exp(\iota s/2)}\mathcal{L}(U_i(t))$$

Using the Pade approximation method, one has

$$\exp(-\iota s)\mathcal{L}(U_i(t)) \approx \frac{(1-\iota s/2)}{(1+\iota s/2)}\mathcal{L}\{U_i(t)\}$$

where s represents the Laplace variable in frequency domain and $\mathcal{L}\{U_i(t)\}$ is the Laplace transform of $U_i(t)$.

For small time delay, the Pade approximation error $\exp(-\iota s) - (1-\iota s/2)/(1+\iota s/2)$ is close to zero. An intermediate variable χ_{2i-1} is introduced and

$$\frac{1-\iota s/2}{1+\iota s/2}\mathcal{L}\{U_i(t)\} = \mathcal{L}\{\chi_{2i-1}(t)\} - \mathcal{L}\{U_i(t)\} \tag{5}$$

Simplifying (5) yields

$$2\mathcal{L}\{U_i(t)\} = \mathcal{L}\{\chi_{2i-1}(t)\} + \frac{\iota s}{2}\mathcal{L}\{\chi_{2i-1}(t)\} \tag{6}$$

The Laplace transform of (6) gives

$$\dot{\chi}_{2i-1} = \frac{4}{\iota}U_i - \frac{2}{\iota}\chi_{2i-1} \tag{7}$$

Equation (7) can be rewritten as

$$\dot{\chi}_{2i-1} = 2\lambda U_i - \lambda\chi_{2i-1} \tag{8}$$

where $\lambda = 2/\iota$.

Finally, the (4) can be transformed as follows:

$$\begin{cases} \dot{x}_{2i-1} = x_{2i} \\ \dot{x}_{2i} = f_i + b_i\Big(\chi_{2i-1} - U_i(t)\Big) \\ \dot{\chi}_{2i-1} = 2\lambda U_i - \lambda\chi_{2i-1} \quad (i = 1, 2, 3, 4, 5, 6) \end{cases} \tag{9}$$

2.1 Multi-Dimensional Taylor Network

The MTN is made up of three layers: the input, middle, and output layers as shown in Fig 2. The middle layer consists of the polynomial combination of inputs, which are easy to adjust. For more details on MTN, the reader is urged to see [40, 41]. Any continuous nonlinear function can be approximated with MTN as follows:

where $s_1, \cdots, s_n$ are the input of MTN.

$$f_{\mathrm{MTN}}(s) = W^T\Theta(s) \tag{10}$$

where $s = [s_1, \cdots, s_n]^{\mathrm{T}} \in \Omega_n \subset \mathcal{R}^n$ is the input vector, $W = [w_1, \cdots, w_l]^{\mathrm{T}} \subset \mathcal{R}^l$ is the weight vector, and $\Theta(s)$

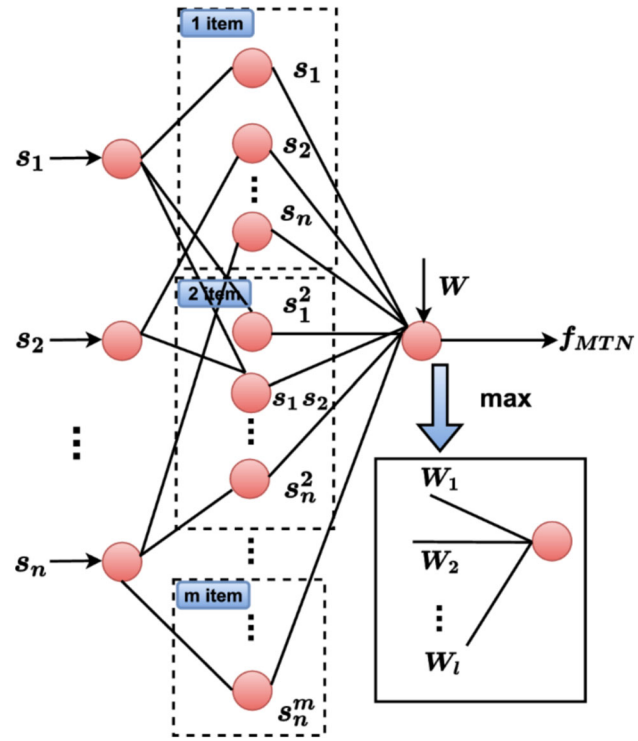

Fig. 2 The architecture of MTN [41]

is the vector of the polynomial combination of inputs and is constructed as follows:

$$\Theta(s) = [\underbrace{s_1, \cdots, s_n}_{1\text{ term}}, \underbrace{s_1^2, \cdots, s_n^2}_{2\text{ term}}, \cdots, \underbrace{s_1^m, \cdots, s_n^m}_{m\text{ term}}]^{\mathrm{T}} \subset \mathcal{R}^l \tag{11}$$

where n is the dimension vector of the input layer and ll is the dimension of the middle layer.

Remark 1 Each element of the vector $\Theta(s)$ is given by $\prod_{i,j=1}^{n} s_i^{\sigma_i} s_j^{\sigma_j}$, where σ_i and σ_j are nonnegative integers and $1 \le \sigma_i + \sigma_j \le m$ holds.

Lemma 1 *[44] For any variable y and constants ε, $0 < \gamma < 1$, and $0 < p < 1$, the following relation holds:*

$$-y(y+\varepsilon)^{\gamma} \le -\frac{1-p}{1+\gamma}y^{1+\gamma} + \frac{\check{\varepsilon}}{1+\gamma} \tag{12}$$

where

$$\check{\varepsilon} = \varepsilon^{1+\gamma} + \left(\frac{\varepsilon}{1-(1-p)^{1/(1+\gamma)}}\right)^{1+\gamma} + \left(\frac{\varepsilon(1-p)^{1/(1+\gamma)}}{1-(1-p)^{1/1+\gamma}}\right)^{1+\gamma} \tag{13}$$

Lemma 2 *[45] For any nonlinear system $\dot{x} = F(x)$, if there exists positive constants $n_1 > 0, n_2 > 0, n_3 > 0$, and $0 < \gamma < 1$, and a Lyapunov function $L(x)$ satisfying $\dot{L}(x) \leq -n_1 L(x) - n_2 V^{\gamma}(x) + n_3$, then the nonlinear system $\dot{x} = F(x)$ is finite-time stable and $L(x)$ converges to the following residual set:*

$$\Upsilon = \left\{ x \mid L(x) \leq \min \left\{ \frac{n_3}{n_1(1-\varsigma)}, \left(\frac{n_3}{n_2(1-\varsigma)} \right)^{\frac{1}{\gamma}} \right\} \right\} \tag{14}$$

The convergence time T_c is given by:

$$T_c \leq \max \left\{ t_0 + \frac{1}{\varsigma n_1 (1-\gamma)} \ln \frac{\varsigma n_1 V^{1-\gamma}(t_0) + n_2}{n_2}, \; t_0 + \frac{1}{n_1(1-\gamma)} \ln \frac{n_1 V^{1-\gamma}(t_0) + \varsigma n_2}{\varsigma n_2} \right\} \tag{15}$$

where $0 < \varsigma_i < 1$ is a scalar.

3 Quadrotor Control Design

This section presents the design of the FTADSC for the quadrotor to guarantee finite-time convergence to the desired states $x_{2i-1,d}$ $(i = 1, 2, 3, 4, 5, 6)$ in the presence of input delay and external disturbances. The whole control structure is depicted in Fig. 3.

Define the tracking error of an ith subsystem as

$$e_{2i-1} = x_{2i-1} - x_{2i-1,d} \quad (i = 1, 2, 3, 4, 5, 6) \tag{16}$$

Then, the derivative of e_{2i-1} with respect to time gives

$$\begin{aligned} \dot{e}_{2i-1} &= \dot{x}_{2i-1} - \dot{x}_{2i-1,d} \\ &= x_{2i} - \dot{x}_{2i-1,d} \end{aligned} \tag{17}$$

The tracking error between x_{2i} and the virtual controller, $x_{2i,d}$, is

$$e_{2i} = x_{2i} - x_{2i,d} + \frac{b_i}{\lambda} \chi_{2i-1} \quad (i = 1, 2, 3, 4, 5, 6) \tag{18}$$

Remark 2 The term $\frac{b_i}{\lambda}\chi_{2i-1}$ is added to compensate for the input delay similar to the approach in [42].

From (18), we have $x_{2i} = e_{2i} + x_{2i,d} - \frac{b_i}{\lambda}\chi_{2i-1}$ and (17) yields

$$\dot{e}_{2i-1} = e_{2i} + x_{2i,d} - \frac{b_i}{\lambda}\chi_{2i-1} - \dot{x}_{2i-1,d} \tag{19}$$

We chose the first positive Lyapunov function as follows:

$$L_{2i-1} = \frac{1}{2} e_{2i-1}^2 \quad (i = 1, 2, 3, 4, 5, 6) \tag{20}$$

$$\begin{aligned} \dot{L}_{2i-1} &= e_{2i-1}\dot{e}_{2i-1} = e_{2i-1}(e_{2i} + x_{2i,d} \\ &\quad - \frac{b_i}{\lambda}\chi_{2i-1} - \dot{x}_{2i-1,d}) \end{aligned} \tag{21}$$

The virtual controller is thus chosen as

$$x_{2i,d} = -c_{2i-1}e_{2i-1} - \kappa_{2i-1}e_{2i-1}^{\gamma} + \dot{x}_{2i-1,d} \tag{22}$$

where c_{2i-1} and κ_{2i-1} are positive constants. (21) is semi-negative definite using (22)

$$\begin{aligned} \dot{L}_{2i-1} &= -c_{2i-1}e_{2i-1}^2 - \kappa_{2i-1}e_{2i-1}^{\gamma+1} \\ &\quad + e_{2i-1}e_{2i} - e_{2i-1}\frac{b_i}{\lambda}\chi_{2i-1} \end{aligned} \tag{23}$$

The time derivative of e_{2i} in (18) gives

$$\begin{aligned} \dot{e}_{2i} &= \dot{x}_{2i} - \dot{x}_{2i,d} \\ &= f_i + b_i\Big(\chi_{2i-1} - U_i(t)\Big) + \delta_i - \dot{x}_{2i,d} + \frac{b_i}{\lambda}\dot{\chi}_{2i} \end{aligned} \tag{24}$$

To eliminate the input delay, (8) is substituted into (24) as follows:

$$\begin{aligned} \dot{e}_{2i} &= f_i + b_i\Big(\chi_{2i-1} - U_i(t)\Big) + \delta_i - \dot{x}_{2i,d} \\ &\quad + \frac{b_i}{\lambda}(2\lambda U_i - \lambda\chi_{2i-1}) \\ &= f_i + b_i U_i(t) + \delta_i - \dot{x}_{2i,d} \end{aligned} \tag{25}$$

To avoid the "explosion of complexity" associated with $\dot{x}_{2i,d}$ in (25), $x_{2i,d}$ is passed through a modified first-order filter to obtain the filtered virtual controller $\hat{x}_{2i,d}$ as follows:

$$\begin{aligned} \varepsilon_i\dot{\hat{x}}_{2i,d} + \hat{x}_{2i,d} &= x_{2i,d} - \varepsilon_i\rho_i^{\gamma} - \varepsilon_i\hat{M}_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right) - \varepsilon_i e_{2i} \\ x_{2i,d}(0) &= \hat{x}_{2i,d}(0) \quad (i = 1, 2, 3, 4, 5, 6) \end{aligned} \tag{26}$$

$$\dot{\hat{M}}_i = \varepsilon_i\left[\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right) - \tau_{2i}\hat{M}_i - \bar{\tau}_{2i}\hat{M}_i^{\gamma}\right] \tag{27}$$

where $\varepsilon_i > 0$ is a constant, $\rho_i = \hat{x}_{2i,d} - x_{2i,d}$ is the boundary layer error, and $\hat{M}_i$ is the estimate of M_i. M_i is the upper bound of $|\dot{x}_{2i,d}|$ which is the effect of differentiating the boundary layer error.

The time derivative of the boundary layer error $\rho_i = \hat{x}_{2i,d} - x_{2i,d}$ is computed as

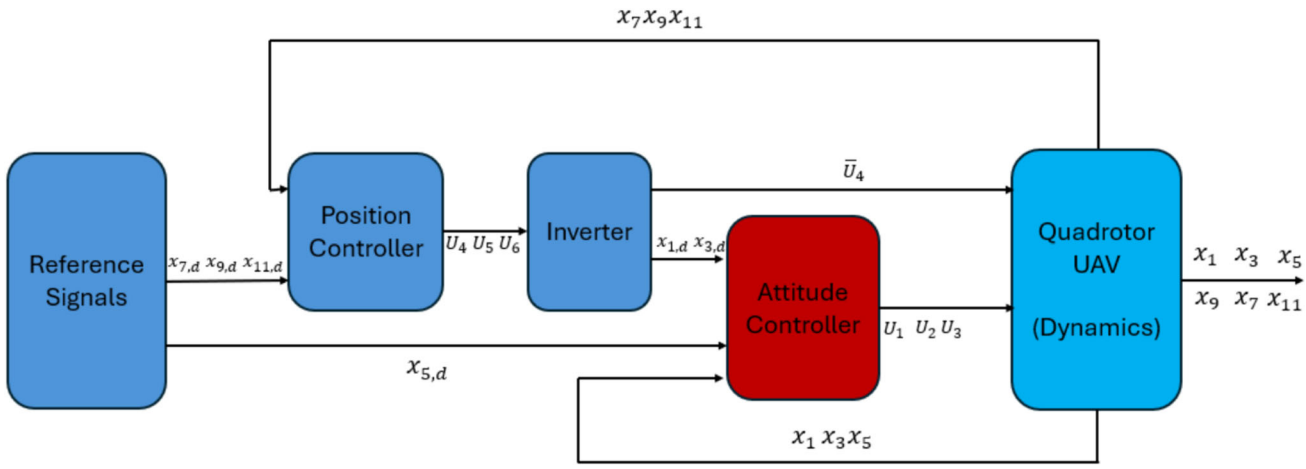

Fig. 3 Block diagram of the quadrotor control system

$$\begin{aligned}\dot{\rho}_i &= \dot{\hat{x}}_{2i,d} - \dot{x}_{2i,d}\\ &= \frac{\hat{x}_{2i,d} - x_{2i,d}}{\varepsilon_i} - \rho_i^{\gamma} - \hat{M}_i \tanh\left(\frac{\rho_i}{\varepsilon_i}\right) - e_{2i}\\ &+ \Upsilon_i\left(e_{2i-1}, e_{2i-1}^{\gamma}, x_{2i-1}, \dot{x}_{2i-1}, x_{2i-1,d}, \dot{x}_{2i-1,d}\right)\\ &= -\frac{\rho_i}{\varepsilon_i} - \rho_i^{\gamma} - e_{2i} - \hat{M}_i \tanh\left(\frac{\rho_i}{\varepsilon_i}\right) + M_i \end{aligned} \tag{28}$$

where $\Upsilon_i(*) = -\dot{x}_{2i,d}$ appears as the effect of differentiating ρ_i, $|\Upsilon_i(*)| \le M_i$, with M_i being the unknown upper bound of $\Upsilon_i(*)$. The modified filter will compensate for M_i through the adaptive parameter $\hat{M}_i$ in finite time.

The second Lyapunov function is selected as

$$L_{2i} = \frac{1}{2}e_{2i-1}^2 + \frac{1}{2}e_{2i}^2 + \frac{\rho_i^2}{2} + \frac{\tilde{\beta}_i^2}{2\alpha_i} + \frac{\tilde{c}_{2i}^2}{2\eta_i} + \frac{\tilde{\kappa}_{2i}^2}{2\mu_i} + \frac{\tilde{M}_i^2}{2\varepsilon_i} \tag{29}$$

where $\tilde{\beta}_i = \hat{\beta}_i - \beta_i$, $\tilde{\kappa}_{2i} = \hat{\kappa}_{2i} - \kappa_{2i}$, and $\tilde{c}_{2i} = \hat{c}_{2i} - c_{2i}$ are estimation errors, $\hat{\kappa}_{2i}$ and $\hat{c}_{2i}$ are adaptive parameters of the controller, $\hat{\beta}_i$ is an adaptive parameter for adjusting the MTN, and $\alpha_i > 0$, $\mu_i > 0$, and $\eta_i > 0$ are design constants. The time derivative of L_{2i} with respect to time yields

$$\begin{aligned}\dot{L}_{2i} &= e_{2i-1}\dot{e}_{2i-1} + e_{2i}\dot{e}_{2i} + \rho_i\dot{\rho}_i + \frac{\tilde{\beta}_i}{\alpha_i}\dot{\tilde{\beta}}_i + \frac{\tilde{c}_{2i}}{\eta_i}\dot{\tilde{c}}_{2i}\\ &+ \frac{\tilde{\kappa}_{2i}}{\mu_i}\dot{\tilde{\kappa}}_{2i} + \frac{\tilde{M}_i}{\varepsilon_i}\dot{\tilde{M}}_i\\ &= -c_{2i-1}e_{2i-1}^2 - \kappa_{2i-1}e_{2i-1}^{\gamma+1} + e_{2i-1}e_{2i} - \frac{b_i}{\lambda}e_{2i-1}\chi_{2i-1}\\ &+ e_{2i}(f_i + b_iU_i(t) + \delta_i - \dot{x}_{2i,d})\\ &- \frac{\rho_i^2}{\varepsilon_i} - \rho_i^{\gamma+1} - \rho_i e_{2i} + \rho_i M_i - \rho_i\hat{M}_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right)\\ &+ \frac{\tilde{\beta}_i}{\alpha_i}\dot{\hat{\beta}}_i + \frac{\tilde{c}_{2i}}{\eta_i}\dot{\hat{c}}_{2i} + \frac{\tilde{\kappa}_{2i}}{\mu_i}\dot{\hat{\kappa}}_{2i} + \frac{\tilde{M}_i}{\varepsilon_i}\dot{\hat{M}}_i \end{aligned} \tag{30}$$

From (30), we know that

$$\begin{aligned}\rho_i M_i - \rho_i\hat{M}_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right) &= \rho_i M_i\\ &- \rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right)\left[\tilde{M}_i + M_i\right]\\ &= M_i\rho_i - M_i\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right)\\ &- \tilde{M}_i\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right) \end{aligned} \tag{31}$$

where $\tilde{M}_i = \hat{M}_i - M_i$ is the estimation error. According to Lemma, $M_i\left[\rho_i - \rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right)\right] \le 0.2785\varepsilon_i M_i$. Then, (31) can be rewritten as

$$\rho_i M_i - \rho_i\hat{M}_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right) \le -\tilde{M}_i\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right) + 0.2785\varepsilon_i M_i \tag{32}$$

Using (32), (30) becomes

$$\begin{aligned}\dot{L}_{2i} &= -c_{2i-1}e_{2i-1}^2 - \kappa_{2i-1}e_{2i-1}^{\gamma+1} - \frac{\rho_i^2}{\varepsilon_i} - \rho_i^{\gamma+1} - \rho_i e_{2i}\\ &+ e_{2i}\left(e_{2i-1} + f_i + \delta_i - \frac{b_i}{\lambda}\frac{e_{2i-1}}{e_{2i}}\chi_{2i-1} + b_iU_i - \dot{x}_{2i,d}\right)\\ &- \tilde{M}_i\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right) + 0.2785\varepsilon_i M_i\\ &+ \frac{\tilde{\beta}_i}{\alpha_i}\dot{\hat{\beta}}_i + \frac{\tilde{c}_{2i}}{\eta_i}\dot{\hat{c}}_{2i} + \frac{\tilde{\kappa}_{2i}}{\mu_i}\dot{\hat{\kappa}}_{2i} + \frac{\tilde{M}_i}{\varepsilon_i}\dot{\hat{M}}_i \end{aligned} \tag{33}$$

The nonlinear terms $f_i+\delta_i-\frac{b_i}{\lambda}\frac{e_{2i-1}}{e_{2i}}\chi_{2i-1}$ can be approximated with the aid of neural network as follows:

$$f_i+\delta_i-\frac{b_i}{\lambda}\frac{e_{2i-1}}{e_{2i}}\chi_{2i-1}=W_i^T\Theta_i(Y_i)+\sigma_i(Y_i) \tag{34}$$

Inserting (34) into (33), one has

$$\begin{aligned}\dot{L}_{2i}&=-c_{2i-1}e_{2i-1}^2-\kappa_{2i-1}e_{2i-1}^{\gamma+1}-\frac{\rho_i^2}{\varepsilon_i}-\rho_i^{\gamma+1}-\rho_i e_{2i}\\&+e_{2i}\left(e_{2i-1}+W_i^T\Theta_i(Y_i)+\sigma_i(Y_i)+b_iU_i-\dot{x}_{2i,d}\right)\\&-\tilde{M}_i\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right)+0.2785\varepsilon_iM_i\\&+\frac{\tilde{\beta}_i}{\alpha_i}\dot{\hat{\beta}}_i+\frac{\tilde{c}_{2i}}{\eta_i}\dot{\hat{c}}_{2i}+\frac{\tilde{\kappa}_{2i}}{\mu_i}\dot{\hat{\kappa}}_{2i}+\frac{\tilde{M}_i}{\varepsilon_i}\dot{\hat{M}}_i\end{aligned} \tag{35}$$

Considering Young's inequalities, we get

$$e_{2i}W_i^T\Theta_i(Y_i)\le\frac{e_{2i}^2W_i^TW_i\Theta_i(Y_i)^T\Theta_i(Y_i)}{2}+\frac{1}{2} \tag{36}$$

Remark 3 Define a scalar $\beta_i=W_i^TW_i=\|W_i\|$ to be the norm of the weight vector of the neural network. The scalar $\hat{\beta}_i$ will be updated online instead of the weight vector to simplify the MTN further.

Using the inequality (36), (35) can be rewritten as follows:

$$\begin{aligned}\dot{L}_{2i}&=-c_{2i-1}e_{2i-1}^2-\kappa_{2i-1}e_{2i-1}^{\gamma+1}-\frac{\rho_i^2}{\varepsilon_i}-\rho_i^{\gamma+1}-\rho_i e_{2i}\\&+e_{2i}\left(e_{2i-1}+e_{2i}\beta_i\Theta_i^T\Theta_i+\sigma_i+b_iU_i-\dot{x}_{2i,d}\right)\\&-\tilde{M}_i\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right)+0.2785\varepsilon_iM_i\\&+\frac{\tilde{\beta}_i}{\alpha_i}\dot{\hat{\beta}}_i+\frac{\tilde{c}_{2i}}{\eta_i}\dot{\hat{c}}_{2i}+\frac{\tilde{\kappa}_{2i}}{\mu_i}\dot{\hat{\kappa}}_{2i}+\frac{\tilde{M}_i}{\varepsilon_i}\dot{\hat{M}}_i\end{aligned} \tag{37}$$

The control input for an ith subsystem of the quadrotor is designed as

$$U_i=\left(-e_{2i-1}-e_{2i}\hat{\beta}_i\Theta_i^T\Theta_i-\hat{c}_{2i}e_{2i}-\hat{\kappa}_{2i}e_{2i}^{\gamma}+\dot{\hat{x}}_{2i,d}\right)/bi \tag{38}$$

The parameters of each ith control input above are updated online using the following adaptive laws.

$$\begin{cases}\dot{\hat{c}}_{2i}=\eta_i\left[e_{2i}^2-\tau_{2i}\hat{c}_{2i}-\bar{\tau}_{2i}\hat{c}_{2i}^{\gamma}\right]\\\dot{\hat{\beta}}_i=\alpha_i\left[e_{2i}^2\Theta_i^T\Theta_i-\tau_{2i}\hat{\beta}_i-\bar{\tau}_{2i}\hat{\beta}_i^{\gamma}\right]\\\dot{\hat{\kappa}}_{2i}=\mu_i\left[e_{2i}^{\gamma+1}-\tau_{2i}\hat{\kappa}_{2i}-\bar{\tau}_{2i}\hat{\kappa}_{2i}^{\gamma}\right]\end{cases} \tag{39}$$

where τ_{2i} and $\bar{\tau}_{2i}$ are design constants. Inserting (38) into (37) gives

$$\begin{aligned}\dot{L}_{2i}&\le-c_{2i-1}e_{2i-1}^2-\kappa_{2i-1}e_{2i-1}^{\gamma+1}-\frac{\rho_i^2}{\varepsilon_i}-\rho_i^{\gamma+1}-\rho_ie_{2i}\\&+e_{2i}\left(-e_{2i}\tilde{\beta}_i\Theta_i^T\Theta_i-(c_{2i}+\tilde{c}_{2i})e_{2i}-(\kappa_{2i}+\tilde{\kappa}_{2i})e_{2i}^{\gamma}+\rho_i\right)\\&-\tilde{M}_i\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right)+\frac{\tilde{\beta}_i}{\alpha_i}\dot{\hat{\beta}}_i+\frac{\tilde{c}_{2i}}{\eta_i}\dot{\hat{c}}_{2i}+\frac{\tilde{M}_i}{\varepsilon_i}\dot{\hat{M}}_i+\frac{\tilde{\kappa}_{2i}}{\mu_i}\dot{\hat{\kappa}}_{2i}\\&+0.2785\varepsilon_iM_i+e_{2i}\sigma_i\\&\le-c_{2i-1}e_{2i-1}^2-\kappa_{2i-1}e_{2i-1}^{\gamma+1}-\frac{\rho_i^2}{\varepsilon_i}-\rho_i^{\gamma+1}-c_{2i}e_{2i}^2-\kappa_{2i}e_{2i}^{\gamma+1}\\&-e_{2i}\rho_i+e_{2i}\rho_i-\tilde{\beta}_i\left(e_{2i}^2\Theta_i^T\Theta_i-\frac{\dot{\hat{\beta}}_i}{\alpha_i}\right)-\tilde{c}_{2i}\left(e_{2i}^2-\frac{\dot{\hat{c}}_{2i}}{\eta_i}\right)\\&-\tilde{\kappa}_{2i}\left(e_{2i}^{\gamma+1}-\frac{\dot{\hat{\kappa}}_{2i}}{\mu_i}\right)-\tilde{M}_i\left(\rho_i\tanh\left(\frac{\rho_i}{\varepsilon_i}\right)-\frac{\dot{\hat{M}}_i}{\varepsilon_i}\right)\\&+e_{2i}\sigma_i+0.2785\varepsilon_iM_i\end{aligned} \tag{40}$$

Theorem 1 *For the quadrotor UAV (9) under the proposed DSC input (38) with a modified filter (26) and estimation laws given by (27) and (39), its closed-loop system is ultimately bounded in a finite time.*

Proof Using (27) and (39), (40) becomes

$$\begin{aligned}\dot{L}_{2i}&\le-c_{2i-1}e_{2i-1}^2-\tau_{2i}e_{2i-1}^{\gamma+1}-\frac{\rho_i^2}{\varepsilon_i}-\rho_i^{\gamma+1}-c_{2i}e_{2i}^2-\kappa_{2i}e_{2i}^{\gamma+1}\\&-\tau_{2i}\tilde{\beta}_i\hat{\beta}_i-\tau_{2i}\tilde{c}_{2i}\hat{c}_{2i}-\tau_{2i}\tilde{\kappa}_{2i}\hat{\kappa}_{2i}-\tau_{2i}\tilde{M}_i\hat{M}_i\\&-\bar{\tau}_{2i}\tilde{\beta}_i\hat{\beta}_i^{\gamma}-\bar{\tau}_{2i}\tilde{c}_{2i}\hat{c}_{2i}^{\gamma}-\bar{\tau}_{2i}\tilde{\kappa}_{2i}\hat{\kappa}_{2i}^{\gamma}-\bar{\tau}_{2i}\tilde{M}_i\hat{M}_i^{\gamma}\\&+e_{2i}\sigma_i+0.2785\varepsilon_iM_i\end{aligned} \tag{41}$$

Using the Young's inequalities, we achieve the following relations

$$\begin{cases}-\tilde{\beta}_i\hat{\beta}_i=-(\beta_i+\tilde{\beta}_i)\tilde{\beta}_i\le\frac{\beta_i^2}{2}-\frac{\tilde{\beta}_i^2}{2}\\-\tilde{\kappa}_{2i}\hat{\kappa}_{2i}=-(\kappa_{2i}+\tilde{\kappa}_{2i})\tilde{\kappa}_{2i}\le\frac{\kappa_{2i}^2}{2}-\frac{\tilde{\kappa}_{2i}^2}{2}\\-\tilde{M}_i\hat{M}_i=-(M_i+\tilde{M}_i)\tilde{M}_i\le\frac{M_i^2}{2}-\frac{\tilde{M}_i^2}{2}\\-\tilde{c}_{2i}\hat{c}_{2i}=-(c_{2i}+\tilde{c}_{2i})\tilde{c}_{2i}\le\frac{c_{2i}^2}{2}-\frac{\tilde{c}_{2i}^2}{2};\\e_2\sigma_i\le\frac{e_{2i}^2}{2}+\frac{\sigma_i^2}{2}\end{cases} \tag{42}$$

From Lemma 1, we achieve the following inequalities

$$\begin{cases}-\tilde{\beta}_i\hat{\beta}_i^{\gamma}=-\tilde{\beta}_i(\tilde{\beta}_i+\beta_i)^{\gamma}\le-\frac{1-p}{1+\gamma}\tilde{\beta}_i^{1+\gamma}+\frac{\check{\beta}_i}{1+\gamma}\\-\tilde{c}_{2i}\hat{c}_{2i}^{\gamma}=-\tilde{c}_{2i}(\tilde{c}_{2i}+c_{2i-1})^{\gamma}\le-\frac{1-p}{1+\gamma}\tilde{c}_{2i}^{1+\gamma}+\frac{\check{c}_{2i}}{1+\gamma}\\-\tilde{\kappa}_{2i}\hat{\kappa}_{2i}^{\gamma}=-\tilde{\kappa}_{2i}(\kappa_{2i}+\tilde{\kappa}_{2i})^{\gamma}\le-\frac{1-p}{1+\gamma}\tilde{\kappa}_{2i}^{1+\gamma}+\frac{\check{\kappa}_{2i}}{1+\gamma}\\-\tilde{M}_i\hat{M}_i^{\gamma}=-\tilde{M}_i(M_i+\tilde{M}_i)^{\gamma}\le-\frac{1-p}{1+\gamma}\tilde{M}_i^{1+\gamma}+\frac{\check{M}_i}{1+\gamma}\end{cases} \tag{43}$$

Table 1 Control design parameters

Parameters	Values	Parameters	Values
c_1, c_3, c_5	7,7,7	c_7, c_9, c_{11}	9,9,9
$\kappa_1, \kappa_3, \kappa_5$	0.5,0.5,0.5	$\kappa_7, \kappa_9, \kappa_{11}$	0.5,0.5,0.5
$\varepsilon_1, \varepsilon_2, \varepsilon_3$	0.2,0.2,0.2	$\varepsilon_4, \varepsilon_5, \varepsilon_6$	0.2,0.2,0.2
τ_1, τ_2, τ_3	0.01,0.01,0.01	τ_4, τ_5, τ_6	0.01,0.01,0.01
$\bar{\tau}_1, \bar{\tau}_2, \bar{\tau}_3$	0.01,0.01,0.01	$\bar{\tau}_4, \bar{\tau}_5, \bar{\tau}_6$	0.01,0.01,0.01
η_1, η_2, η_3	15, 15, 15	η_4, η_5, η_6	26, 26, 26
μ_1, μ_2, μ_3	1.4, 1.4, 1.4	μ_4, μ_5, μ_6	3, 3, 3
$\alpha_1, \alpha_2, \alpha_3$	0.012, 0.012, 0.012	$\alpha_4, \alpha_5, \alpha_6$	0.017, 0.017, 0.017
γ	0.6		

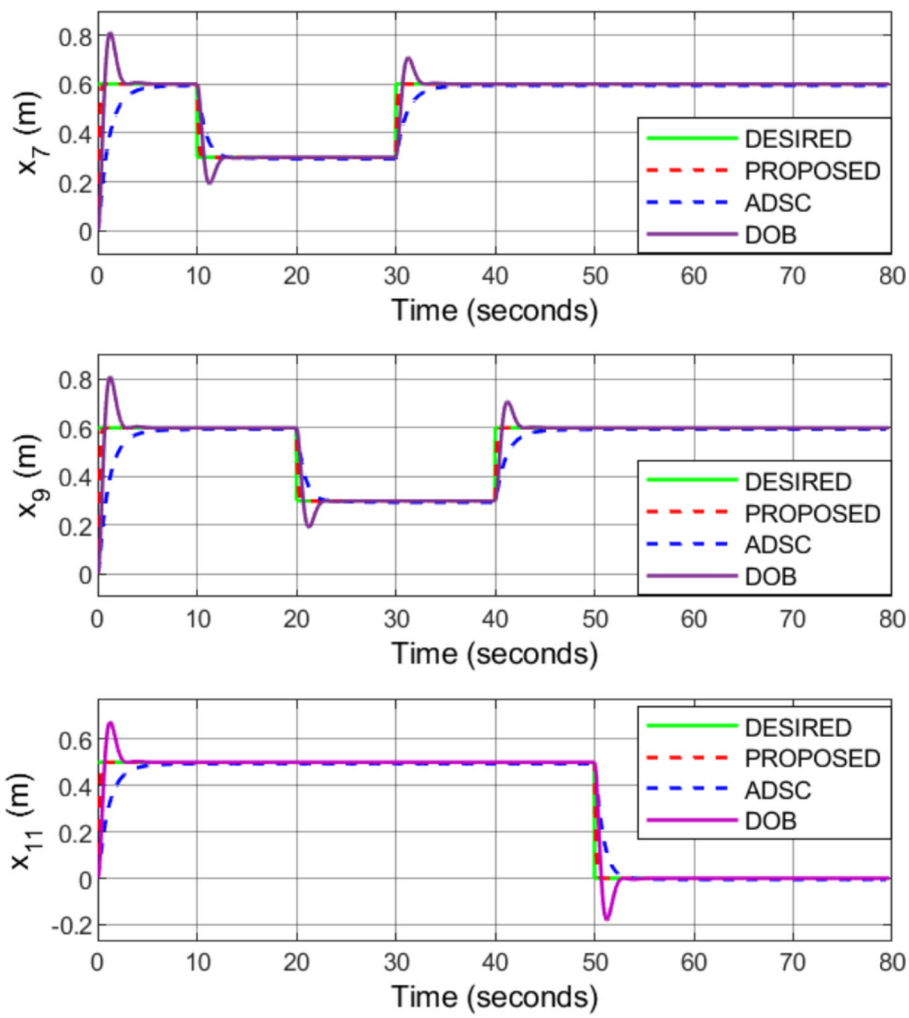

Fig. 4 Position response for x_7, x_9, and x_{11}

Substituting the inequalities (42) and (43) into (41), one has

$$\dot{L}_{2i} \leq -c_{2i-1}e_{2i-1}^2 - \kappa_{2i-1}\left(e_{2i-1}^2\right)^{\frac{\gamma+1}{2}} - \left(c_{2i} - \frac{1}{2}\right)e_{2i}^2 - \kappa_{2i}\left(e_{2i}^2\right)^{\frac{\gamma+1}{2}} - \left(\frac{1}{\varepsilon_i}\right)\rho_i^2 - \left(\rho_i^2\right)^{\frac{\gamma+1}{2}} - \tau_{2i}\frac{\tilde{\beta}_i^2}{2} - \bar{\tau}_{2i}\frac{1-p}{1+\gamma}\frac{(\tilde{\beta}_i^2)^{\frac{\gamma+1}{2}}}{2} - \tau_{2i}\frac{\tilde{c}_{2i}^2}{2} - \bar{\tau}_{2i}\frac{1-p}{1+\gamma}\frac{(\tilde{c}_{2i}^2)^{\frac{\gamma+1}{2}}}{2} - \tau_{2i}\frac{\tilde{\kappa}_{2i}^2}{2} - \bar{\tau}_{2i}\frac{1-p}{1+\gamma}\frac{\left(\tilde{\kappa}_{2i}^2\right)^{\frac{1+\gamma}{2}}}{2} - \tau_{2i}\frac{\tilde{M}_i^2}{2} - \bar{\tau}_{2i}\frac{1-p}{1+\gamma}\frac{\left(\tilde{M}_i^2\right)^{\frac{1+\gamma}{2}}}{2} + \tau_{2i}\frac{c_{2i}^2}{2} + \tau_{2i}\frac{a_{2i-1}^2}{2} + \tau_{2i}\frac{\kappa_{2i}^2}{2}$$

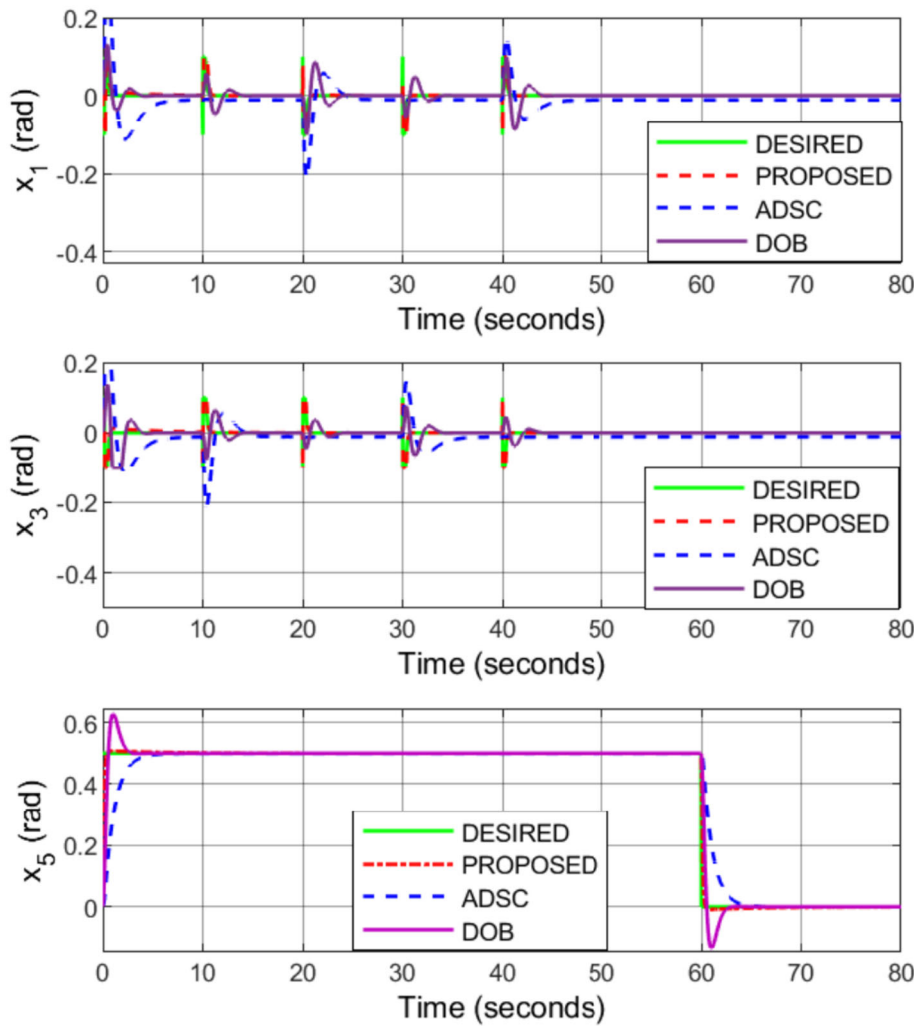

Fig. 5 Attitude angle response for x_1, x_3, and x_5

$$+\tau_{2i}\frac{M_i^2}{2}+\bar{\tau}_{2i}\frac{\check{c}_{2i}}{1+\gamma}+\bar{\tau}_{2i}\frac{\check{\beta}_i}{1+\gamma}+\bar{\tau}_{2i}\frac{\check{M}_i}{1+\gamma}+\bar{\tau}_{2i}\frac{\check{\kappa}_{2i}}{1+\gamma}$$
$$+0.2785\varepsilon_i M_i+\frac{\sigma_i^2}{2} \tag{44}$$

Let

$$r_{2i}=min\left\{2c_{2i-1},\ 2\left(c_{2i}-\frac{1}{2}\right),\ 2\left(\frac{1}{\varepsilon_i}\right),\ \tau_{2i},\ \tau_{2i},\ \tau_{2i},\ \tau_{2i}\right\},$$
$$q_{2i}=min\Big\{\kappa_{2i-1}2^{\frac{\gamma+1}{2}},\kappa_{2i}2^{\frac{\gamma+1}{2}},2^{\frac{\gamma+1}{2}},$$
$$\frac{\bar{\tau}_{2i}}{2}\frac{1-p}{1+\gamma}.2^{\frac{\gamma+1}{2}},\frac{\bar{\tau}_{2i}}{2}\frac{1-p}{1+\gamma}.2^{\frac{\gamma+1}{2}},$$
$$\frac{\bar{\tau}_{2i}}{2}\frac{1-p}{1+\gamma}.2^{\frac{\gamma+1}{2}},\frac{\bar{\tau}_{2i}}{2}\frac{1-p}{1+\gamma}.2^{\frac{\gamma+1}{2}}\Big\}$$

$o_{2i}=\tau_{2i}\frac{c_{2i}^2}{2}+\tau_{2i}\frac{a_{2i-1}^2}{2}+\tau_{2i}\frac{\kappa_{2i}^2}{2}+\tau_{2i}\frac{M_i^2}{2}+\bar{\tau}_{2i}\frac{\check{c}_{2i}}{1+\gamma}+\bar{\tau}_{2i}\frac{\check{\beta}_i}{1+\gamma}+\bar{\tau}_{2i}\frac{\check{M}_i}{1+\gamma}+\bar{\tau}_{2i}\frac{\check{\kappa}_{2i}}{1+\gamma}+0.2785\varepsilon_i M_i+\frac{\sigma_i^2}{2}$. Then, (44) can be simplified as follows:

$$\dot{L}_{2i}=-r_{2i}L_{2i}-q_{2i}L_{2i}^{\frac{\gamma+1}{2}}+o_{2i} \tag{45}$$

According to Lemma 2, the Lyapunov function L_{2i} converges to the set $L_{2i}\leq\min\left\{\frac{o_{2i}}{r_{2i}(1-\varsigma)},\left(\frac{o_{2i}}{q_{2i}(1-\varsigma)}\right)^{\frac{2}{1+\gamma}}\right\}$ with a convergence time of

$$T_c\leq\max\left\{t_0+\frac{1}{\varsigma r_{2i}\left(1-\frac{\gamma+1}{2}\right)}\ln\frac{\varsigma r_{2i}V^{1-\frac{\gamma+1}{2}}(t_0)+q_{2i}}{q_{2i}},\right.$$
$$t_0+\frac{1}{r_{2i}\left(1-\frac{\gamma+1}{2}\right)}\ln$$
$$\left.\frac{r_{2i}V^{1-\frac{\gamma+1}{2}}(t_0)+\varsigma q_{2i}}{\varsigma q_{2i}}\right\}$$

□

The Quadrotor motion along x_7 and x_9 directions are determined by the pitch and roll inclinations, respectively. Therefore, the desired roll and pitch orientations are auto-

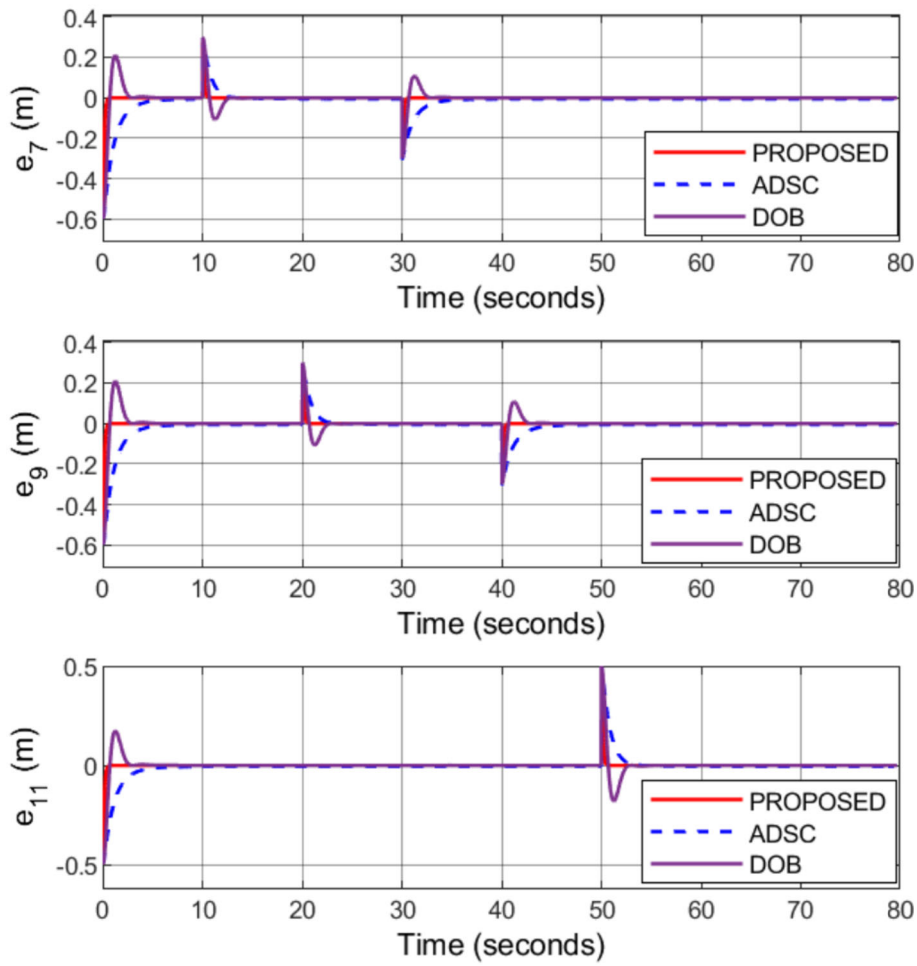

Fig. 6 Tracking errors for the position

matically tuned by the following equations:

$$\begin{cases} x_{1,d} = \arcsin\left(S_{x_{5,d}} U_4 - C_{x_{5,d}} U_5\right) \\ x_{3,d} = \arcsin\left(\frac{U_4 - S_{x_{3,d}} \; S_{x_{5,d}}}{C_{x_{5,d}} \; C_{x_{3,d}}}\right) \end{cases} \tag{46}$$

4 Simulation Results

For the sake of illustrating the efficacy of the proposed control for the quadrotor with input delay, simulations and comparisons are carried out. The proposed method is compared with adaptive DSC (ADSC) designed for an uncertain quadrotor in [17], and disturbance observer-based backstepping (DOB) designed for a quadrotor with input delay in [24].

The quadrotor parameters are obtained from [13]. The initial condition of each quadrotor state is 0.001. The initial condition of each adaptive gain of the proposed controller is 0.001. The initial condition of each adaptive parameter of the MTN is set as 0.05. To set the middle layer of the MTN, we set $n = 2$ and $m = 2$ according to (11), and the input vector is chosen as $s_i = [e_{2i-1} \; x_{2i-1}]^T$. The design parameters of the controllers are provided in Table 1. The input delay is chosen as $\iota = 0.03$. The desired trajectories of the quadrotor are given as follows:

$$x_{7,d} = \begin{cases} \frac{3}{5} & 0 \le t \le 10 \\ \frac{3}{10} & 10 < t \le 30 \\ \frac{3}{5} & 30 < t \le 60 \end{cases} \; ; \; x_{9,d} = \begin{cases} \frac{3}{5} & 0 \le t \le 20 \\ \frac{3}{10} & 20 < t \le 40 \\ \frac{3}{5} & 40 < t \le 60 \end{cases}$$

$$x_{11,d} = \begin{cases} \frac{1}{2} & 0 \le t \le 50 \\ 0 & 50 < t \le 60 \end{cases} \; ; \; x_{5,d} = \begin{cases} \frac{1}{2} & 0 \le t \le 60 \\ 0 & 20 < t \le 30 \end{cases}$$

A parametric uncertainty of $\pm 30\%$ is added to the system and the time-varying external disturbances are modeled as

$$\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} = \begin{bmatrix} 2cos(2t) \\ 2sin(0.61t) \\ 2sin(t) \end{bmatrix}$$

The simulation results are demonstrated in Figs. 4, 5, 6, 7, 8, 9, 10, 11, and 12. From Figs. 4 and 5, it can be seen that the proposed control scheme, ADSC, and DOB have tracked the desired position and attitude trajectories in the

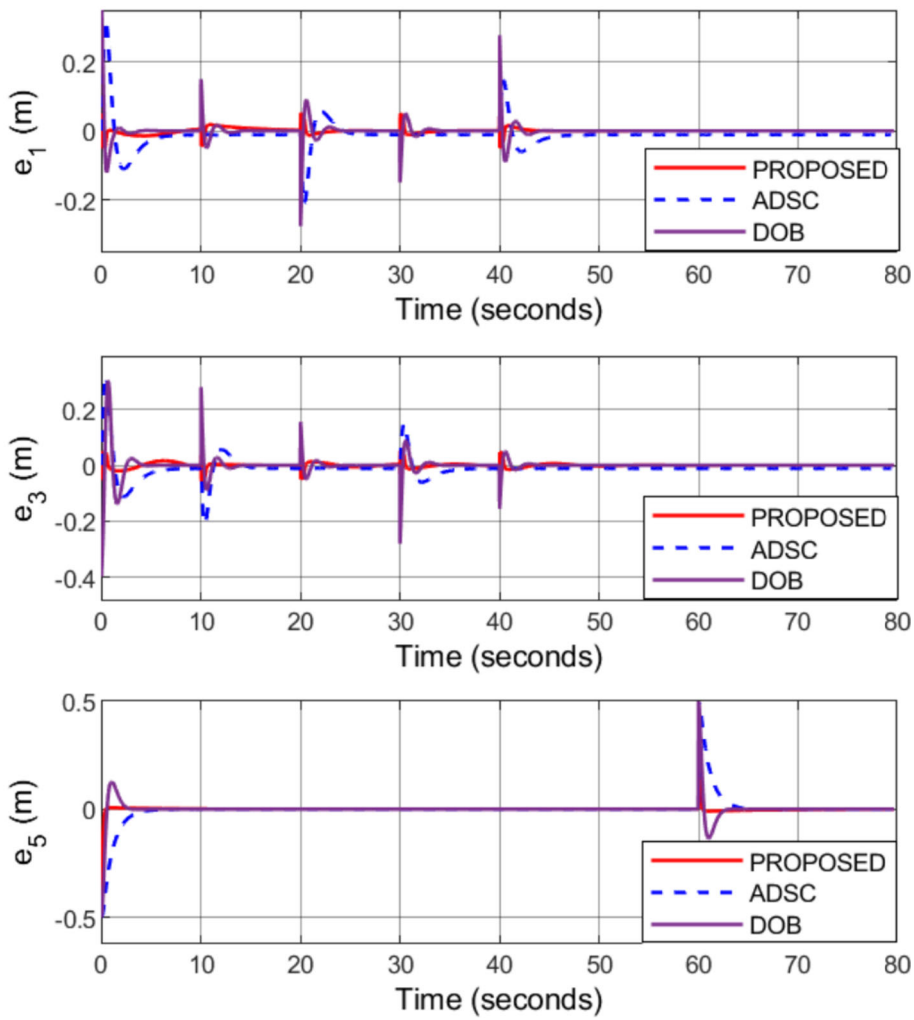

Fig. 7 Tracking errors for the attitude angles

Table 2 Settling time of the three control techniques

Variables	Proposed (s)	DOB (s)	ADSC (s)
x_1	0.91	4.30	6.11
x_3	0.91	4.30	6.11
x_5	0.615	3.12	5.22
x_7	0.80	2.96	5.37
x_9	0.80	2.96	5.37
x_{11}	0.80	2.96	5.37

Table 3 RMSE of the three control techniques

Variables	Proposed	DOB	ADSC
x_1	0.0087	0.0248	0.0403
x_3	0.0094	0.0350	0.0410
x_5	0.0216	0.0402	0.0622
x_7	0.0202	0.0510	0.0650
x_9	0.0206	0.0451	0.0626
x_{11}	0.0194	0.0490	0.0577

Table 4 Maximum overshoots of the three control techniques

Variables (%)	Proposed (%)	DOB (%)	ADSC (%)
x_1	0.0	31.1	99.4
x_3	0.0	45.7	85.0
x_5	0.0	24.6	0.0
x_7	0.0	33.5	0.0
x_9	0.0	33.9	0.0
x_{11}	0.0	37.1	0.0

presence of input delay, parametric variations, and environmental disturbances. Although the DOB can compensate for the input delay, its asymptotic convergence property and the presence of overshoots make its responses slower in tracking the desired trajectories. The ADSC could neither neutralize the effects of the input delay nor guarantee finite-time convergence, which resulted in slow responses. On the other hand, the proposed control scheme not only eliminates the input delay but also self-adjusts its gains in finite time to enhance

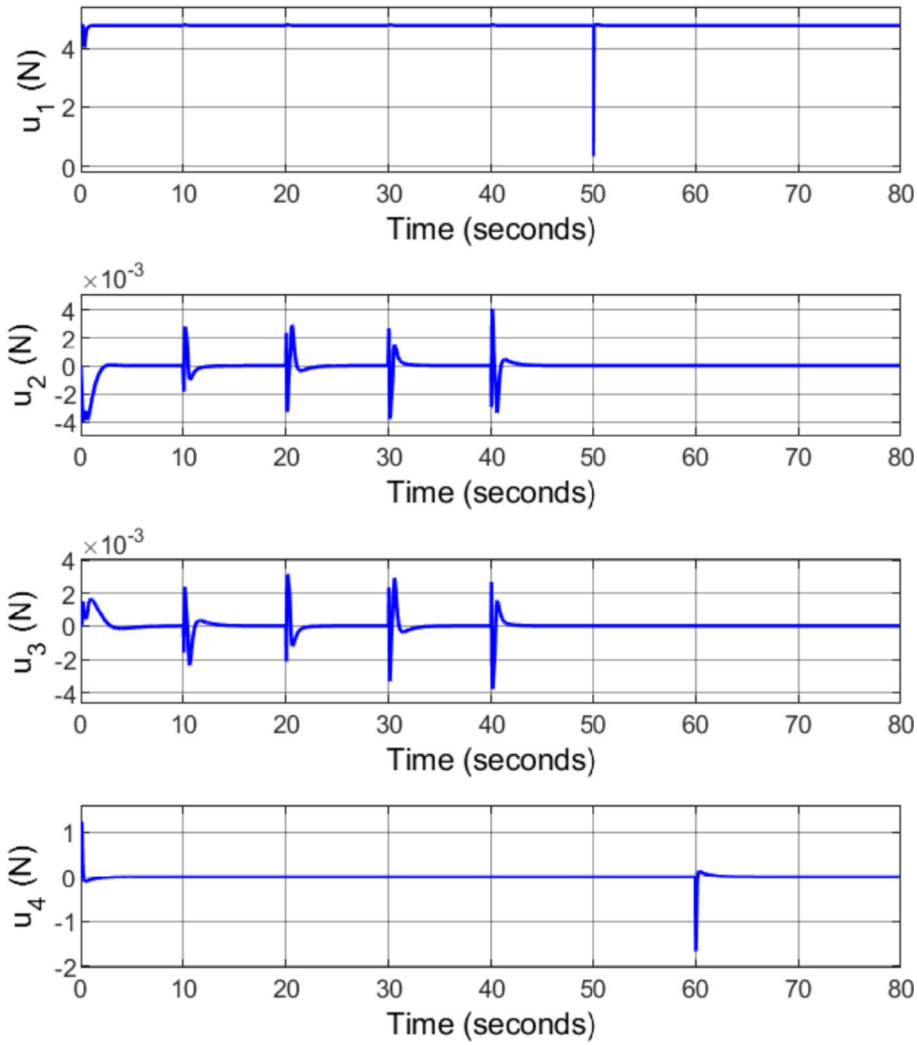

Fig. 8 Quadrotor control signals

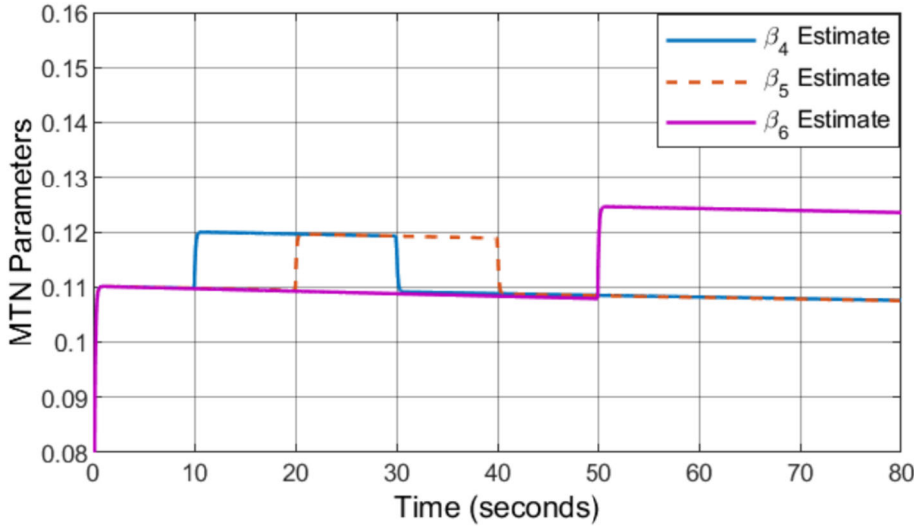

Fig. 9 The MTN parameters for the position coordinates

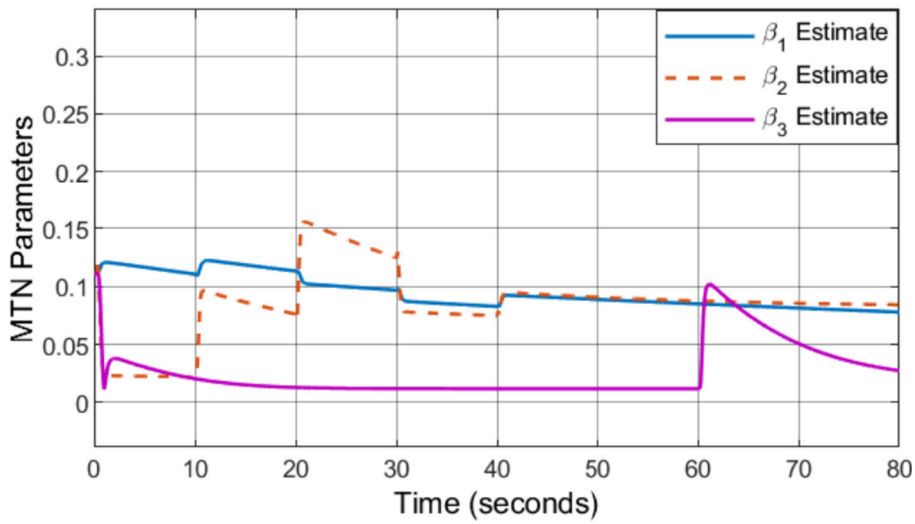

Fig. 10 The MTN parameters for the attitude angles

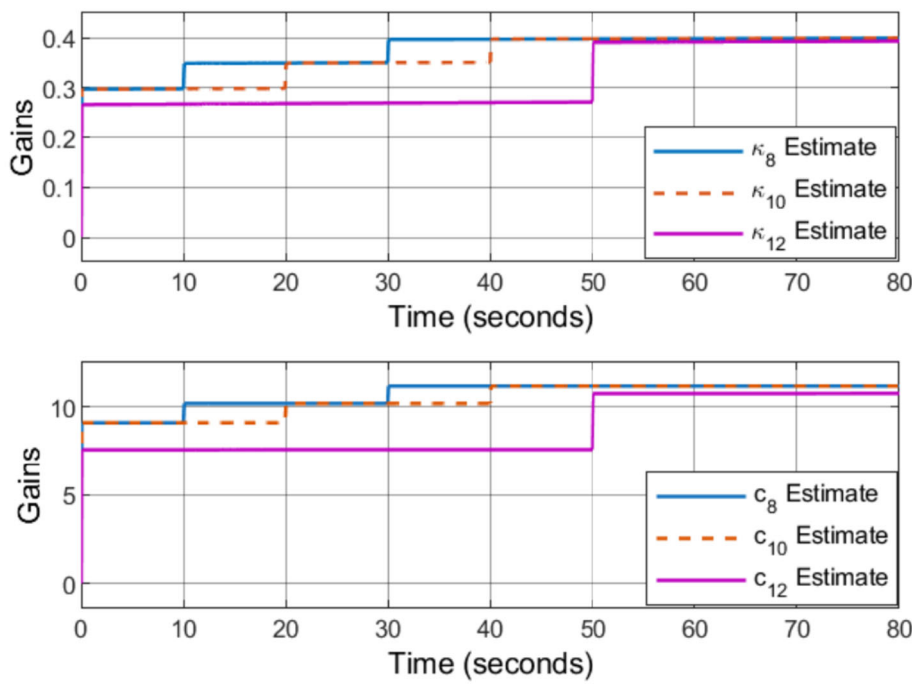

Fig. 11 Adaptive gains of the proposed controller for the position coordinates

tracking accuracy. Another disadvantage of the ADC is its inability to compensate for the filtering error compared to the proposed control scheme. Thus, the proposed control scheme offers the best tracking accuracy with respect to settling time and overshoot as shown in Figs. 4 and 5. The tracking errors for the quadrotor position and attitude under the three strategies are depicted in Figs. 6 and 7, respectively. It can be observed that the position and attitude tracking errors converge to the vicinity of zero faster under the proposed control scheme compared to both ADSC and DOB. The control signals of the quadrotor system that realize the finite-time trajectory tracking with reasonable accuracy are presented in Fig. 8.

The parameters for adjusting the MTN to approximate the uncertain nonlinearities of the quadrotor are presented in Figs. 9 and 10. The estimated gains of the position attitude subsystems of the quadrotor are presented in Figs. 11 and 12, respectively. It is clear that the gains are self-adjusting according to the desired trajectories.

To further illustrate the comparisons of the proposed controller, DOB, and ADSC, Tables 2, 3, and 4 are provided. Table 2 presents the comparison of the settling time of the three controllers. It can be observed that the proposed controller has the smallest settling time despite the input delay. In addition, Table 3 shows that the root-mean-square of the tracking error is smaller under the proposed controller compared to both DOB and ADSC. Moreover, Table

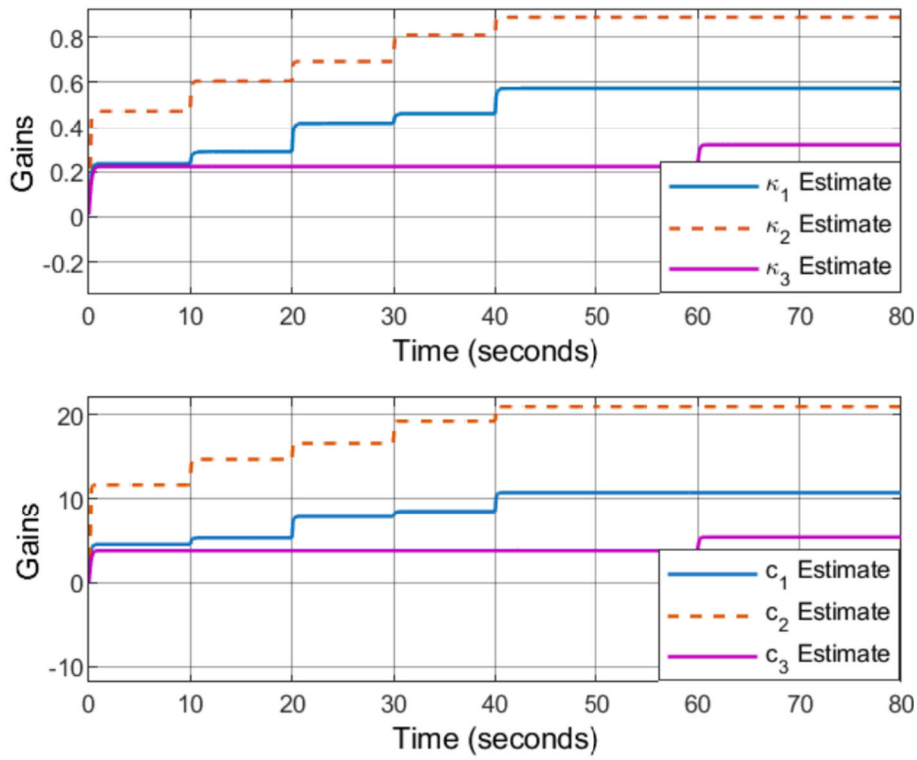

Fig. 12 Adaptive gains of the proposed controller for the attitude angles

4 demonstrates that the maximum overshoot under the proposed controller is the least among the three controllers. Thus, the proposed controller offers the best performance in terms of settling time, root-mean-square of the tracking error, and overshoot for the quadrotor system considered in this study.

5 Conclusions

This paper studied the finite-time adaptive control of a quadrotor with input delay, parametric variations, and external disturbances. The Pade approximation method has been used to eliminate the input delay. A DSC approach realized the attitude and position tracking of the quadrotor in finite time. A modified first-order filter with more terms has been employed to avoid the "explosion of complexity" problem and compensate for the boundary layer errors in finite time. The gains of the DSC are updated online with the aid of adaptive laws so they can self-adjust whenever the target trajectories change to improve tracking accuracy. The uncertain nonlinearities and external disturbances are approximated via the MTN. A single parameter has been used to update the MTN to reduce the computational load. The finite-time convergence of the closed-loop system to a compact set has been proven using the Lyapunov candidate function. The simulations have shown that the proposed control strategy displayed the smallest tracking errors, the least overshoots, and the shortest settling time compared to ADSC and DOB. The remote control of a quadrotor requires a reliable wireless communication. However, this exposes the quadrotor to potential cyber-attacks from adversaries. In the next research, we will investigate the potential of fractional-order DSC and fractional-order observer to mitigate denial-of-service attacks and ensure the quadrotor's trajectory tracking with acceptable accuracy.

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